

QRAFT TECHNOLOGIES, LONG-TERM EQUITY TEAM, 2024

AI Market-Risk Signal for Japan & Taiwan

An ensemble hedging-signal framework for FSITC's First Financial Quantitative Japan Fund — three models, one weekly risk score, a dynamic hedge.

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Problem setting

Developing risk signals for portfolio hedging in the Taiwan and Japan markets

Every week the fund has to answer one question:
how much equity exposure should stay on for the
coming week?

The signal must be early enough to matter, robust
across two very different markets, and simple
enough to execute at Monday's open.



Objective

Weekly risk-on / risk-off signal with a probability, per market



Use

Drives the dynamic hedge ratio of a \$200M quantitative Japan fund



Constraints

Interpretable models, no look-ahead, data by Friday, executed at Monday's open

Two markets, two sets of assumptions

What each market's structure implies for the signal design



Taiwan

- Highly concentrated in semiconductors, with TSMC as the market anchor
- Relatively liquid options market; futures and ETFs widely used
- Performance strongly tied to the global semiconductor cycle

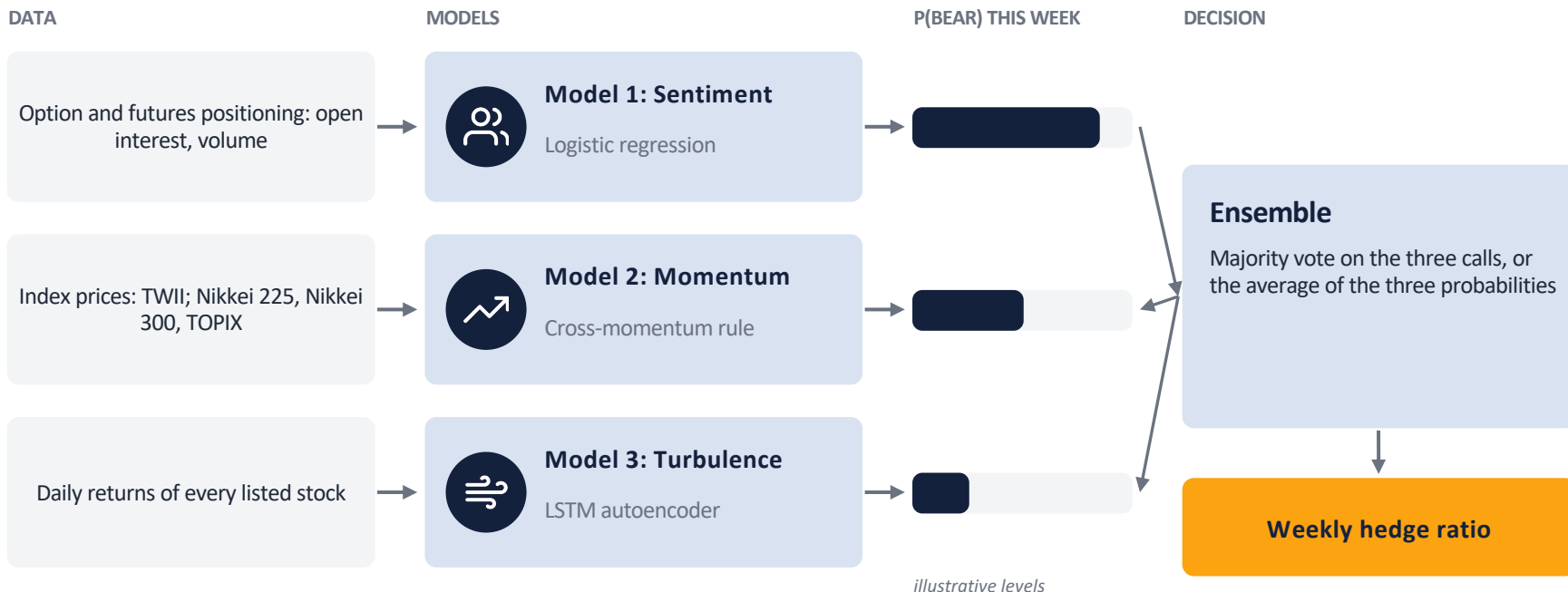


Japan

- Structurally low-volatility market with a well-developed options ecosystem
- Highly sensitive to global macro events through large foreign ownership
- Leadership driven by large caps (Nikkei 225 constituents)

Framework: three perspectives, one ensemble

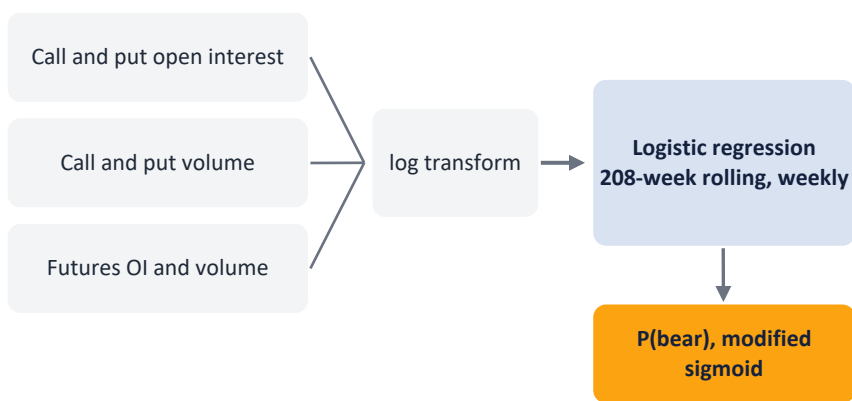
Each model reads a different layer of the market; the ensemble turns three probabilities into one weekly hedge



All three models retrain on a rolling window every week; nothing is fitted on the evaluation period.

Model 1 — Supply & Demand (option positioning)

Investor expectations about market direction, as revealed by option and futures data



Input (log-transformed)	Taiwan	Japan
Call / put open interest	TWSE options	Nikkei options
Call / put trading volume	TWSE options	Nikkei options
Futures open interest & volume	TWSE futures	Nikkei futures
Label	TWSE	Nikkei 225

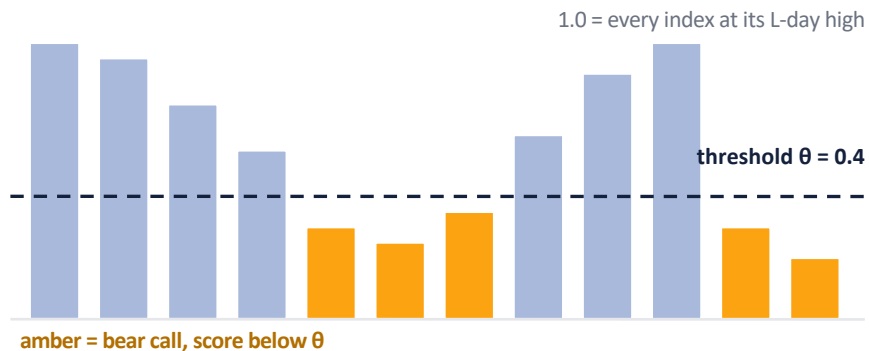
Data source: Bloomberg

- Sudden spikes in implied volatility and put positioning reflect global risk-off sentiment (Li, Chen & French, 2015)
- The classifier predicts the coming week's regime: a binary call plus a probability

Model 2 — Cross-momentum rule

Negative momentum across a market's representative indices is a short-term hedge trigger

CROSS-MOMENTUM SCORE BY DAY (ILLUSTRATIVE)



Example, L = 5	Index 1	Index 2
Price today	99	93
5-day rank	1	3
1 / rank	1.00	0.33
CM score	$(1.00 + 0.33) / 2 = 0.67$	

- Rank today's price within the last L days for each index (TWII; Nikkei 225, Nikkei 300, TOPIX); CM = average of 1 / rank
- Bear when $CM < \theta$; tuned over $L \in \{5, 10, 21\}$ and $\theta \in \{0.2, 0.3, 0.4, 0.5\}$. In low-volatility Japan a sustained momentum shift marks a genuine regime change

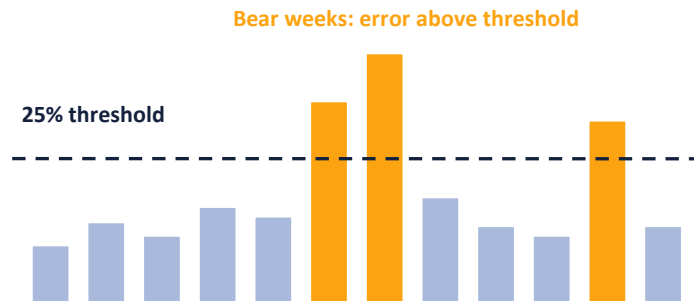
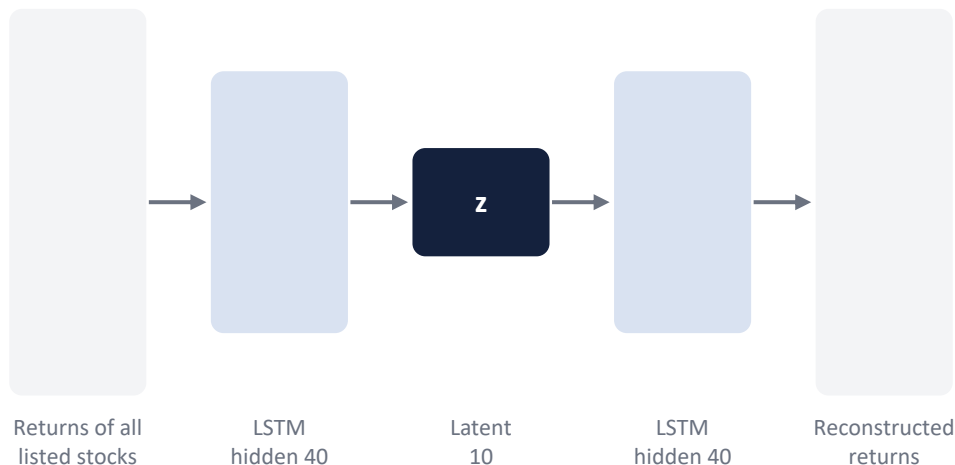
Model 3 — Market turbulence (LSTM autoencoder)

Unsupervised: when the cross-section of stock returns stops looking like the recent past, risk is rising

ENCODER

DECODER

RECONSTRUCTION ERROR BY WEEK (ILLUSTRATIVE)

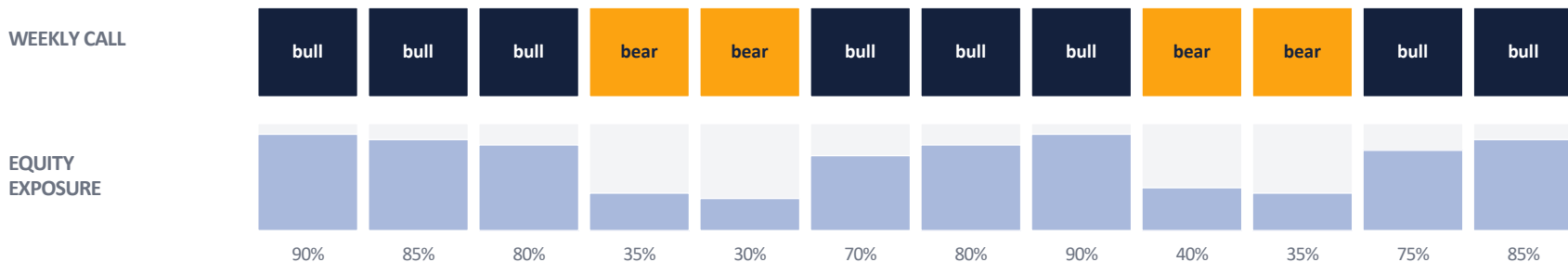


– Universe: TWSE / TSE listed stocks, NaN ratio < 5%; no labels — the model learns the normal cross-sectional structure

– Rolling training 756 days, weekly; inference 5 days. Hidden 40, encoder 10, dropout 0.5, lr 0.001, 10 epochs per step

Ensemble and backtest protocol

Three probabilities become one weekly call and an equity exposure — illustrative 12-week strip



Ensemble rules

- Voting: majority of the three binary calls
- Equal-weight: average of the three probabilities → exposure
- The probability version trades less and gives the headline results

Backtest protocol

- Weekly; data feed by Friday, execution at Monday's open
- 2006-01-03 to 2024-10-11 (981 weeks); costs of 0, 5 and 10 bp one way
- Benchmarks: TWII (Taiwan) and TOPIX (Japan)

Result — Taiwan vs. TWII

Equal-weighted probability ensemble, 5 bp costs, 2006–2024

-22.5% **11.1%**

Max drawdown
(TWII -58.3%)

Annualised vol
(TWII 17.7%)

0.67 **5.0%**

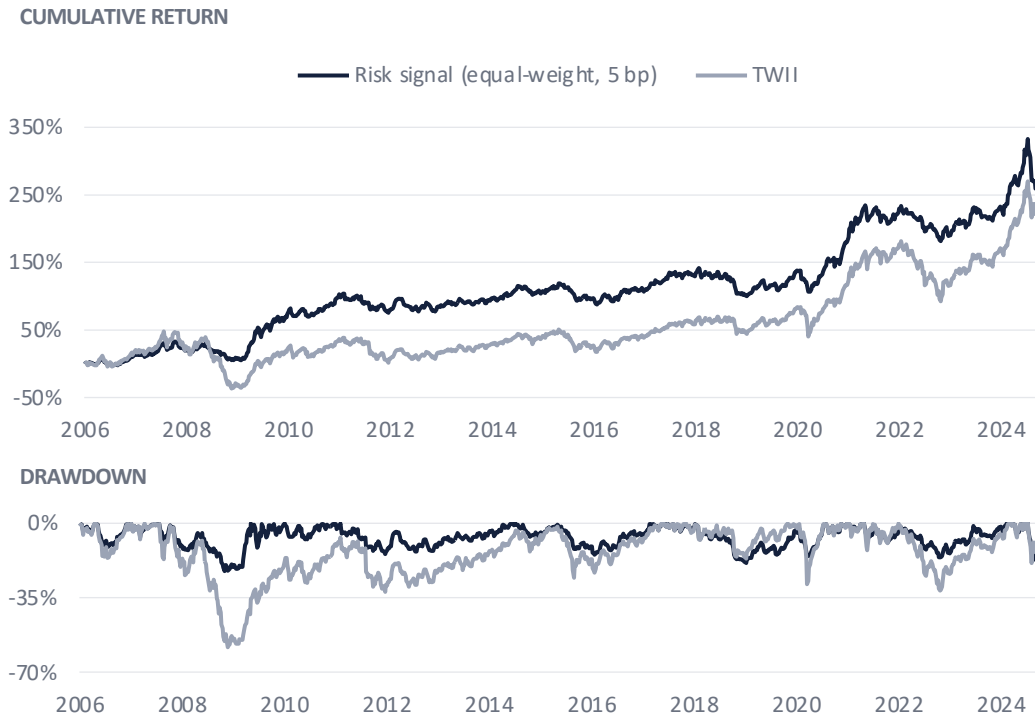
Sharpe
(TWII 0.45)

CAGR
(TWII 4.7%)

278% vs 247%

Cumulative return, signal vs. TWII

Curves from the original backtest report.



Taiwan — every model and ensemble, before costs

Weekly signals 2006–2024 vs. TWII buy-and-hold; binary = hard call, probability = scaled exposure

Variant	Cum. return %	CAGR %	Ann. vol %	Sharpe	Max DD %	Bear weeks
Model 1 Supply & Demand — binary	636.5	7.62	11.79	0.93	−20.84	321 (32%)
Model 1 Supply & Demand — probability	490.3	6.74	10.30	0.94	−16.06	
Model 2 Momentum — binary	269.4	4.92	11.93	0.62	−21.19	351 (35%)
Model 2 Momentum — probability	246.2	4.67	11.49	0.61	−21.03	
Model 3 Turbulence — binary	341.6	5.61	14.31	0.61	−31.63	127 (12%)
Model 3 Turbulence — probability	244.1	4.65	15.35	0.49	−46.92	
Ensemble — voting (binary)	429.0	6.31	12.64	0.74	−24.91	247 (25%)
Ensemble — equal-weight probability	334.3	5.55	11.06	0.74	−22.01	
TWII (benchmark)	247.4	4.68	17.71	0.45	−58.31	—

Every variant beats the index on volatility and drawdown; the positioning model alone is the strongest single signal, and the equal-weight ensemble keeps most of its return with the lowest volatility of the ensembles.

Result — Japan vs. TOPIX

Equal-weighted probability ensemble, 5 bp costs, 2006–2024

-32.7% **11.2%**

Max drawdown
(TOPIX -61.7%)

Annualised vol
(TOPIX 20.8%)

0.27 **1.7%**

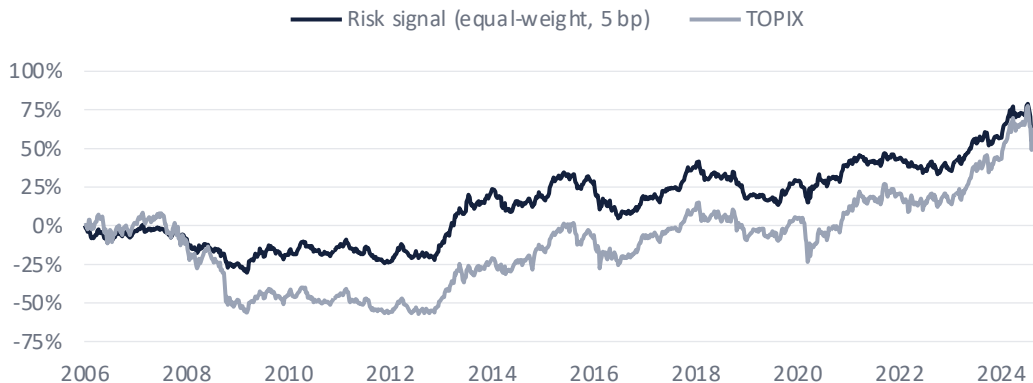
Sharpe
(TOPIX 0.23)

CAGR
(TOPIX 1.8%)

Roughly half the volatility and drawdown of the index at a similar return — the profile a hedging overlay is meant to deliver in a low-return market.

Curves from the original backtest report.

CUMULATIVE RETURN



DRAWDOWN



Japan — every model and ensemble, before costs

Weekly signals 2006–2024 vs. TOPIX buy-and-hold; binary = hard call, probability = scaled exposure

Variant	Cum. return %	CAGR %	Ann. vol %	Sharpe	Max DD %	Bear weeks
Model 1 Supply & Demand — binary	134.0	3.17	12.38	0.42	–28.37	513 (49%)
Model 1 Supply & Demand — probability	110.2	2.77	10.38	0.42	–25.33	
Model 2 Momentum — binary	152.8	3.47	11.50	0.47	–28.92	581 (56%)
Model 2 Momentum — probability	101.2	2.60	10.91	0.38	–26.18	
Model 3 Turbulence — binary	23.3	0.77	15.81	0.15	–49.25	171 (16%)
Model 3 Turbulence — probability	37.1	1.17	18.02	0.18	–53.82	
Ensemble — voting (binary)	141.8	3.30	13.32	0.41	–26.71	373 (36%)
Ensemble — equal-weight probability	89.5	2.38	11.21	0.35	–31.48	
TOPIX (benchmark)	64.0	1.84	20.81	0.23	–61.72	—

In a low-return market the positioning and momentum models roughly double the index return at half the volatility; turbulence alone is weak in Japan, which is why the ensemble matters.

Cost sensitivity — ensembles at 0, 5 and 10 bp one way

The probability ensemble trades less, so it degrades more slowly; headline results use 5 bp

Taiwan vs. TWII

Ensemble, cost	Cum. %	CAGR %	Vol %	Sharpe	MDD %
Voting, 0 bp	429.0	6.31	12.64	0.74	-24.91
Voting, 5 bp	370.1	5.85	12.65	0.69	-25.81
Voting, 10 bp	317.7	5.40	12.66	0.64	-26.69
Equal-weight, 0 bp	334.3	5.55	11.06	0.74	-22.01
Equal-weight, 5 bp	277.6	5.00	11.06	0.67	-22.50
Equal-weight, 10 bp	228.2	4.47	11.07	0.61	-22.99
TWII	247.4	4.68	17.71	0.45	-58.31

Japan vs. TOPIX

Ensemble, cost	Cum. %	CAGR %	Vol %	Sharpe	MDD %
Voting, 0 bp	141.8	3.30	13.32	0.41	-26.71
Voting, 5 bp	86.2	2.31	13.34	0.31	-28.39
Voting, 10 bp	43.4	1.33	13.37	0.21	-32.16
Equal-weight, 0 bp	89.5	2.38	11.21	0.35	-31.48
Equal-weight, 5 bp	59.0	1.72	11.21	0.27	-32.72
Equal-weight, 10 bp	33.3	1.06	11.22	0.19	-33.93
TOPIX	64.0	1.84	20.81	0.23	-61.72

Highlighted rows are the headline configuration used elsewhere in this deck. Japan's binary ensemble is more cost-sensitive because it flips exposure fully; the probability version scales exposure and turns over less.

What it delivered

A weekly, interpretable risk score per market from three complementary models — positioning, momentum and cross-sectional turbulence.

Deployed as the core hedging-signal engine for FSITC's \$200M First Financial Quantitative Japan Fund.

Taiwan: max drawdown -22.5% vs -58.3%; Japan: volatility 11.2% vs 20.8%, both after costs.

The weekly client report on top of the signal is generated by a three-agent LLM workflow (see companion deck).

Python, NumPy, PyTorch, scikit-learn, pandas, LangGraph

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